

Introduction

DS2020 – ARTIFICIAL INTELLIGENCE

NARAYANAN C KRISHNAN

General Information

❑ Course Structure

- 3-0-2 (4 credits)

❑ Scheduled Lecture Timings

- Monday 11.00-11.50am
- Wednesday– 11.00-11.50am
- Friday– 11.00-11.50am
- Compensatory classes on Saturday – 10.00am-12.00pm
 - 5 Saturdays in the entire semester

❑ Scheduled Lab Timings

- Wednesday 2.00-3.30pm

❑ Instructor Office Hours

- By appointment through email

❑ Course information and schedule

- <https://seekayan.github.io/s24ds2020.html>

Tentative Course Schedule

week 1 (2) - Introduction and Intelligent Agents (*Compensatory class on Jan 20th, Saturday*)
week 2 (2) - Problem Solving Agents
week 3 (1) - Search I, **lab 1**
week 4 (3) - Search II, (*Compensatory class on Feb 3, Saturday*)
week 5 (3) - Adversarial Search, **lab 2**
week 6 (2) - Logical Agents and Propositional Logic
week 7 (3) - First Order Logic, **lab 3**
week 8 (3) - Planning (*Compensatory class on March 2nd, Saturday*)
week 9 (1) - Probabilistic Reasoning I **lab 4**
week 10 (3) - Probabilistic Reasoning II
week 11 (1) - Decision Theory, **lab 5**
week 12 (2) - Markov Decision Process I (*Compensatory class on March 30, Saturday*)
week 13 (3) - Markov Decision Process II **lab 6**
week 14 (3) - Reinforcement Learning I, (*Compensatory class on April 20th, Saturday*)
week 15 (3) - Reinforcement Learning II
week 16 (1) - Summary

Tests – 30%

- ❑ Covers material discussed from the previous test till the current week
- ❑ Duration 50 minutes
- ❑ All the 2 quizzes will be considered towards the final grade
- ❑ Will follow the timetable issued by the Academic section

Labs – 30%

- ❑ 6 labs during the semester
 - ❑ Each lab will account for 5% of the overall grade
 - ❑ Some labs may be take-home, and some in person
 - ❑ Labs in Python
-

Grading Scheme

□ Tentative Breakup

- Tests (2) – 30%
- End-semester exam – 40%
- Labs (6) - 30%
- Attendance Bonus - 1%
 - Attendance will be taken for every class and will count towards the bonus points

Honor Code

❑ Unless explicitly stated otherwise:

- Strictly individual effort
- Group discussions at a high level are encouraged
- You are forbidden from trawling the web for answers etc.

❑ Any infraction will be dealt with in severest terms allowed.

❑ I reserve the right to question you with regards to your submission, if I suspect any misconduct.

Reference Material

❑ Course Textbook

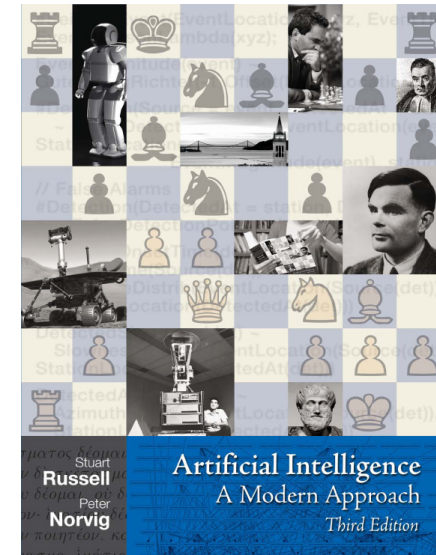
- Artificial Intelligence A Modern Approach, Stuart Russell and Peter Norvig, 3rd edition
- Low price edition will suffice

❑ Other reference materials

- <http://aima.cs.berkeley.edu/>
- AI – Rich and Knight

❑ Pre-requisites

- Probability (for later part of the course)



Course Website

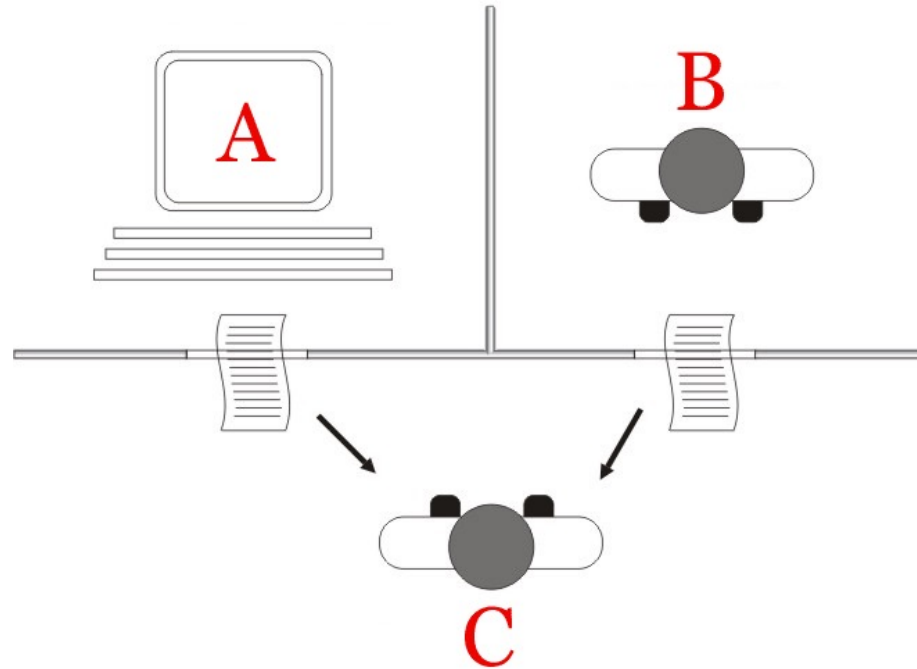
- ❑ Moodle module for the course
- ❑ All class related material will be accessible from the Moodle
- ❑ I will not be giving any separate handouts
- ❑ The pdf version of the lecture slides will be available on Moodle
 - The reading material for the lectures will also be shared on Moodle

Introduction

Motivation – Why study AI?

What comes to your mind when you hear AI?

1950 – Alan Turing – The Turing Test



1955-56 Birth of the word AI

□ John McCarthy

- AI – “the science and engineering of making intelligent machines”

□ 1956 – Dartmouth Summer Research Project on AI



<http://www.dartmouth.edu/~vox/0607/0724/ai50.html>

1964 - Eliza



□ Joseph Weizenbaum

□ Eliza

- Natural Language Processing Computer Program
- First Chatbot!
- DOCTOR - Psychotherapist

```
=====
EEEEEE L      I I I I I 2222222  AAA
E      L      I      Z  A  A
E      L      I      Z  A  A
EEEE   L      I      Z  A  A
E      L      I      Z  A A A A A
E      L      I      Z  A  A
EEEEEE L L L L L I I I I 22222  A  A

=====
ELIZA > (PLEASE TYPE IN ALL CAPS) WHAT'S YOUR NAME DEAR ?
AMIT
ELIZA > HI. I'M ELIZA. WHAT DO YOU WANT TO TALK ABOUT ?
AMIT > CAN I TALK ABOUT MY PROBLEM ?
ELIZA > SURE...
AMIT > DUE TO ON GOING EXAMS IN COLLEGE, I HAVE A LOT OF STRESS.
ELIZA > PLEASE RELAX A LITTLE, SLEEP WELL.
AMIT > THANKS FOR YOUR ADVICE.
ELIZA > NO MENTION.
AMIT > BYE
ELIZA > BYE AND KEEP IN TOUCH...
=====
```

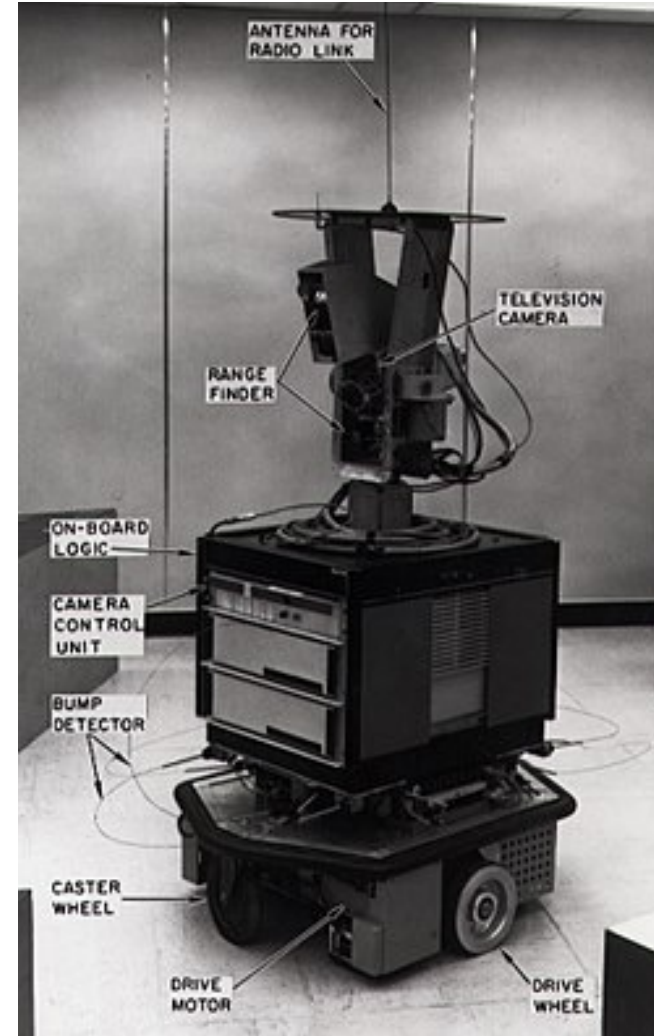
<http://news.mit.edu/2008/obit-weizenbaum-0310>

1966-72 - Shakey

☐ Shakey the robot

- Combined research in robotics, computer vision and natural language processing
- Could reason about its actions!

<http://www.ai.sri.com/shakey/>

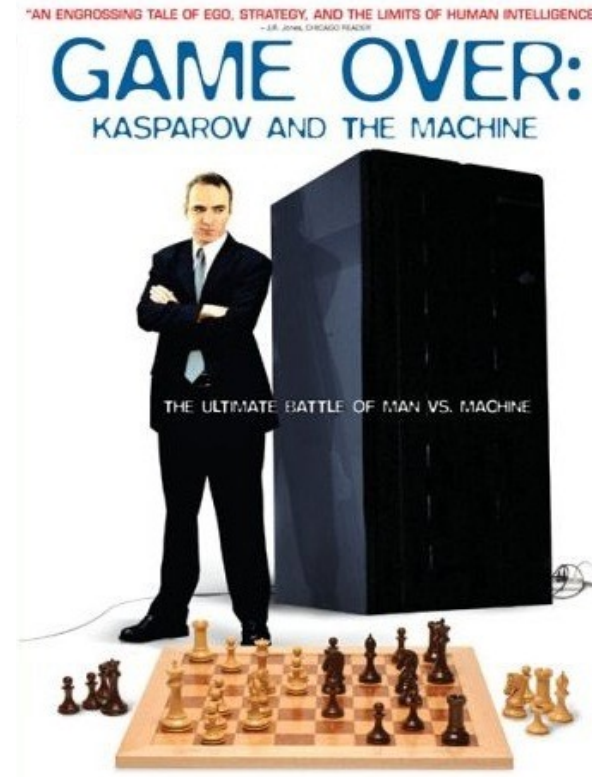
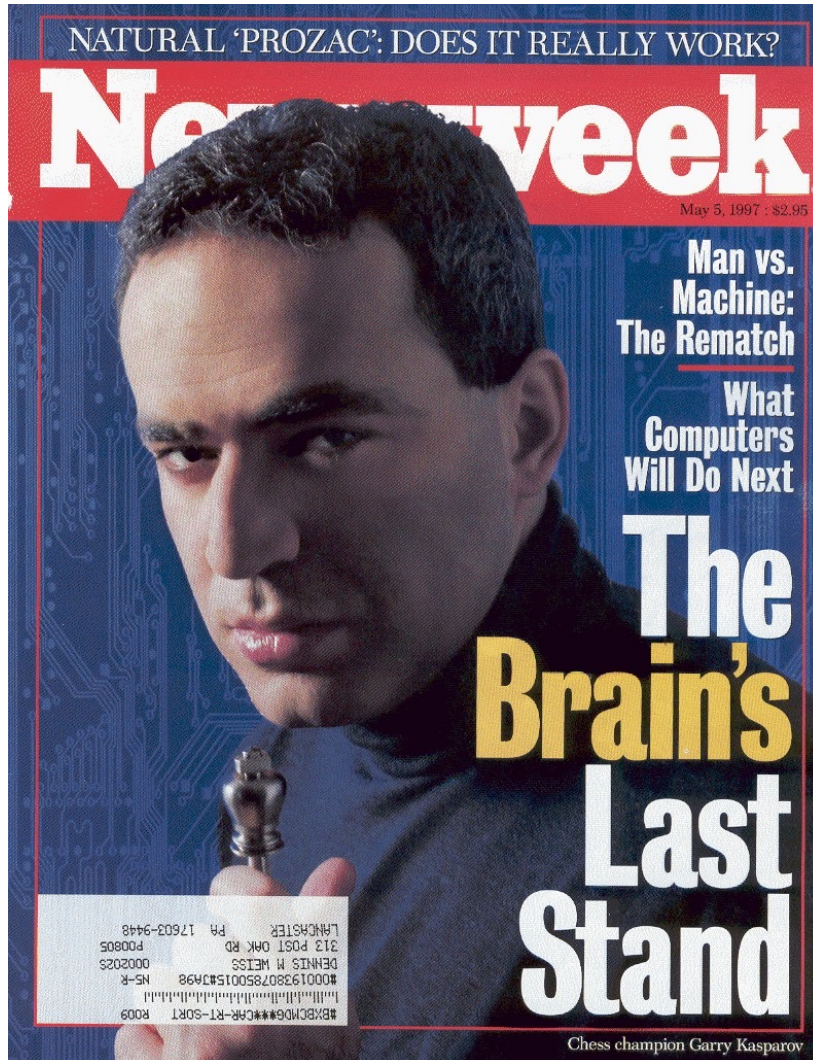


1970-90 – AI Winter

- ❑ Failure of early AI machines for real-world applications
- ❑ Decrease in research funding for AI
- ❑ Expert systems (rule-based systems) bottoming out
- ❑ Neural Networks fail to take off



1997 – IBM Deep Blue



Kasparov said that he sometimes saw **deep intelligence and creativity** in the machine's moves

1997 – First Official Robocup



1999 – SONY AIBO

❑ World's first AI pet!

- Responded to over 100 voice commands and talked back to the human
- Sold out in 20 minutes after launch!!!



2002 – Roomba from iRobot

❑ Autonomous robotic vacuum cleaners

- Avoid obstacles
- Move around corners
- Detect dirty spots on the floor
- Automatic navigation



2004 – DARPA Grand Challenge for Autonomous Driving

❑ Birth of Autonomous Driving



2011 – IBM's Watson

- ❑ Successful demonstration of answering questions posed in natural language
- ❑ Birth of smartphone apps and voice devices

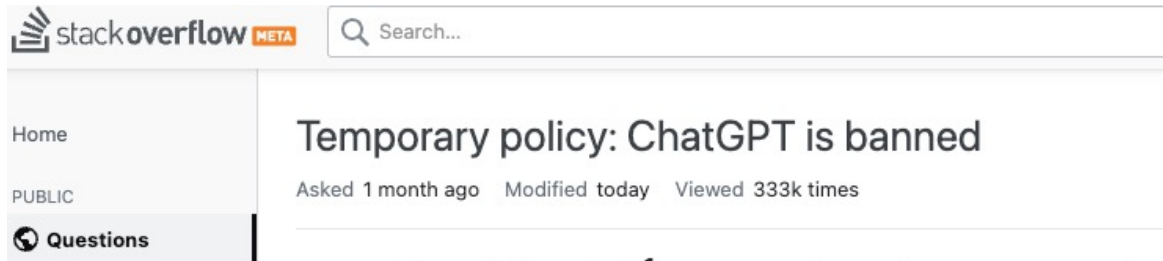


2016 – AI conquers Go!

❑ AI conquers the final frontier in board games



2022 - ChatGPT



▲ For comedic/ironic purposes, I asked ChatGPT "why should stack overflow ban ai generated answers"

1190



Stack Overflow is a website for programmers and developers to ask and answer questions related to coding and computer programming. It is intended to be a resource for people who are looking for help with specific programming problems or who want to learn more about a particular topic. Because AI-generated answers may not always be accurate or relevant, they could potentially cause confusion or mislead users who are looking for help on Stack Overflow. In addition, AI-generated answers may not always follow the formatting and style guidelines of the site, which could make them difficult to read or understand. For these reasons, it may be appropriate for Stack Overflow to ban AI-generated answers.

On the other hand, "why should stack overflow allow ai generated answers?"

There are a few potential benefits to allowing AI-generated answers on Stack Overflow. For one, AI-generated answers could potentially provide more accurate and detailed answers to users' questions. This could be especially useful for complex technical questions that may be difficult for human experts to answer. Additionally, AI-generated answers could help to reduce the workload for human experts, allowing them to focus on more important tasks.

However, there are also some potential drawbacks to consider. For example, AI-generated answers may not always be accurate or helpful, and there may be concerns about the potential for bias in the algorithms used to generate answers. Additionally, some users may have concerns about the use of AI in general, and may prefer to receive answers from human experts.

Overall, whether or not to allow AI-generated answers on Stack Overflow is a complex decision that would need to be carefully considered by the community.



What is AI?

□ What do you think?

Definition of AI

Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

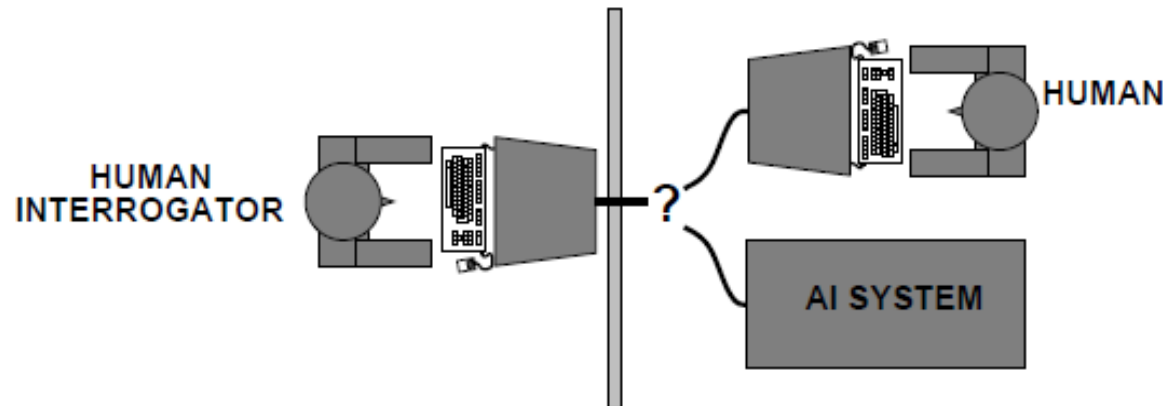
Thinking Humanly “The exciting new effort to make computers think . . . <i>machines with minds</i> , in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents.” (Poole <i>et al.</i> , 1998) “AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

Definition of AI

Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

□ Acting Humanly –

- Turing test



- Is it sufficient to imitate a human (living being)?

Definition of AI

Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

❑ Thinking humanly

- Model human thinking process
- Requires scientific theories of internal activities of the human brain
- Cognitive Science, Neuroscience, Psychology



❑ A machine that thinks like human while solving a problem correctly.

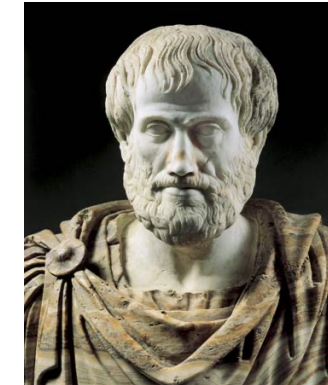
Definition of AI

Thinking humanly	Thinking rationally
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❑ Thinking Rationally

❑ Laws of Thought

- Aristotle – right thinking
- Belief that “logic” governs the human thought process



❑ Knowledge is not always 100% certain

❑ What is the goal? What is purpose of thinking?

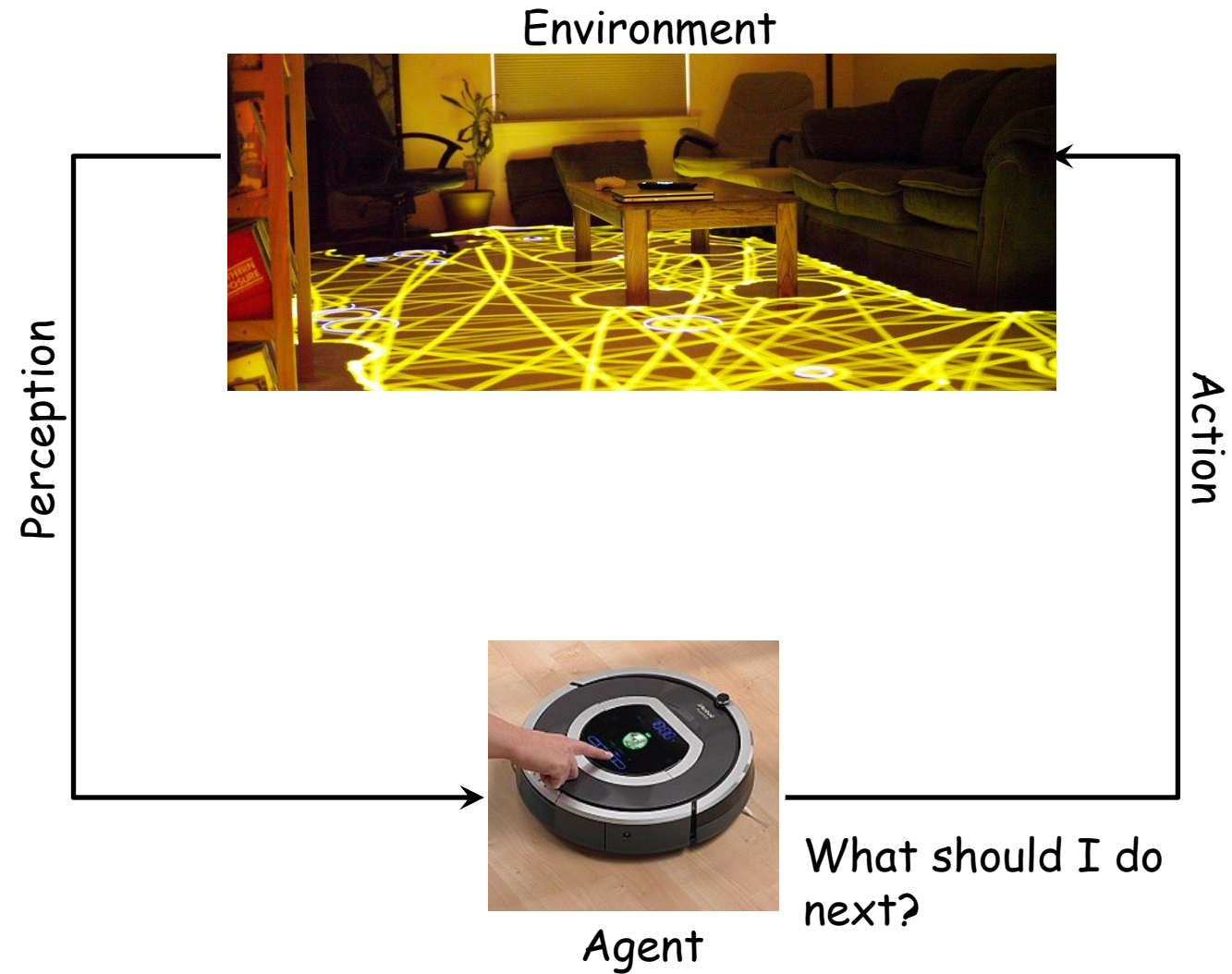
Definition of AI

Thinking humanly	Thinking rationally
Acting humanly	Acting rationally

- ❑ Acting Rationally
- ❑ rational behavior = doing the right thing
- ❑ Agent: an entity that perceives and acts
- ❑ Goal: building rational agents - agents that achieve the best outcome or under uncertainty the expected outcome.
- ❑ Encompasses the other lines of thought.
 - Thinking rationally will help to act rationally, but is not the only means; Eg: Reflex

Intelligent Agents

Agent



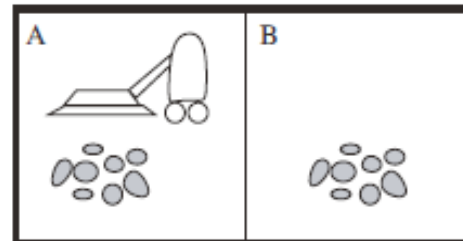
Agent Functions and Program

- ❑ Agent behavior is described by the **agent function** that maps percept sequences to actions.
- ❑ Lookup Table – An action for every possible percept sequence.
- ❑ **Agent Program**: realization/concrete implementation of the agent function within some physical system.

```
function SKELETON-AGENT(percept) returns action
  static: memory, the agent's memory of the world

  memory ← UPDATE-MEMORY(memory, percept)
  action ← CHOOSE-BEST-ACTION(memory)
  memory ← UPDATE-MEMORY(memory, action)
  return action
```

Vacuum World



Percept sequence	Action
$[A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Dirty}]$	<i>Suck</i>
$[B, \textit{Clean}]$	<i>Left</i>
$[B, \textit{Dirty}]$	<i>Suck</i>
$[A, \textit{Clean}], [A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Clean}], [A, \textit{Dirty}]$	<i>Suck</i>
$\vdots$	$\vdots$
$[A, \textit{Clean}], [A, \textit{Clean}], [A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Clean}], [A, \textit{Clean}], [A, \textit{Dirty}]$	<i>Suck</i>
$\vdots$	$\vdots$

Rational Agents

- ❑ A rational agent does the **right thing(action)**
- ❑ Without loss of generality, “goals” specifiable by **performance measure** defining a numerical value for any environment history
- ❑ Rational Action: that maximizes the expected value of the **performance measure** given the percept sequence to date and prior knowledge
- ❑ Rationality $\neq$ Omniscience

PEAS – Specifying the Task Environment

❑ Must specify the task environment as fully as possible

Task Environment for automated taxi driver?

- Performance

- Environment

- Actuator

- Sensors

PEAS – Specifying the Task Environment

❑ Must specify the task environment as fully as possible

Task Environment for automated taxi driver?

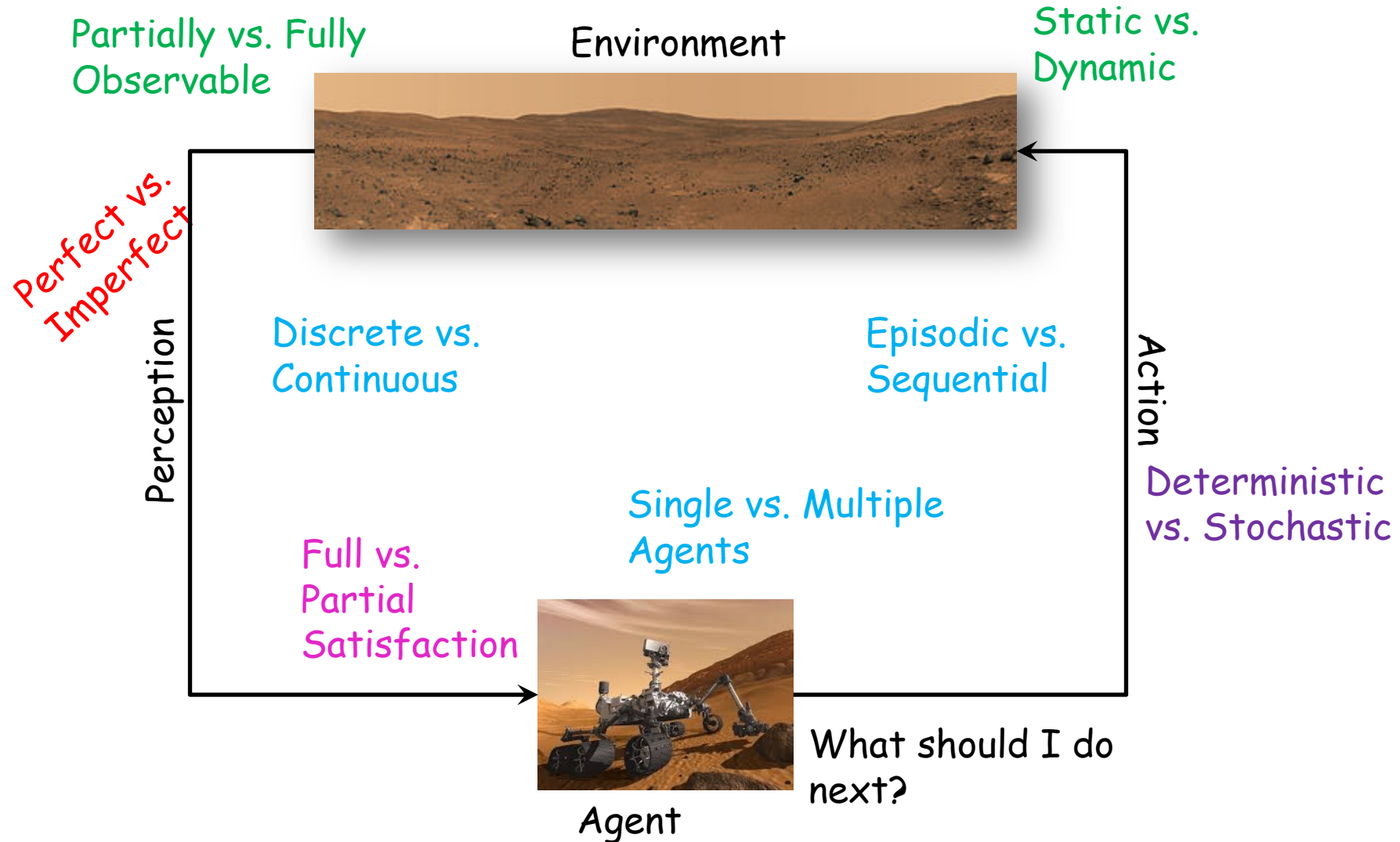
- Performance- safe, fast, comfortable
- Environment-roads, other traffic, traffic signals
- Actuator-steering, accelerator, brake, horn, signal
- Sensors-video camera, IR sensor, GPS, odometer

PEAS – Specifying the Task Environment

❑ How does the following affect the complexity of the problem the rational agent faces?

- Performance – complex goals makes performance harder to achieve?
- Environment
- Actuator – Lack of effectors makes performance harder to achieve?
- Sensors – Lack of percepts makes performance harder to achieve?

Properties of the Task Environment



Properties of the Task Environment

- ❑ **Observable:** The agent can “sense” its environment
 - best: fully observable worst: unobservable typical: partially observable
- ❑ **Deterministic:** The actions have predictable effects
 - best: deterministic worst: non-deterministic typical: stochastic
- ❑ **Static:** The world does not change when the agent is deciding on what to do next
 - best: static worst: dynamic typical: semi-dynamic
- ❑ **Episodic:** The performance of the agent is determined episodically
 - best: episodic worst: non-episodic
- ❑ **Discrete:** The environment evolves through a discrete set of states
 - best: discrete worst: continuous typical: hybrid
- ❑ **Agents:** # of agents in the environment; are they competing or cooperating?

Task Environment-Examples

Environment	Observable	Deterministic	Static	Episodic	Discrete	# Agents
Chess						



Task Environment-Examples

Environment	Observable	Deterministic	Static	Episodic	Discrete	# Agents
Chess	Fully	Deterministic	Semi	Sequential	Discrete	Multi



Task Environment-Examples

Environment	Observable	Deterministic	Static	Episodic	Discrete	# Agents
Chess	Fully	Deterministic	Semi	Sequential	Discrete	Multi
Poker						



Task Environment-Examples

Environment	Observable	Deterministic	Static	Episodic	Discrete	# Agents
Chess	Fully	Deterministic	Static	Sequential	Discrete	Multi
Poker	Partial	Stochastic	Static	Sequential	Discrete	Multi



Task Environment-Examples

Environment	Observable	Deterministic	Static	Episodic	Discrete	# Agents
Chess	Fully	Deterministic	Semi	Sequential	Discrete	Multi
Poker	Partial	Stochastic	Static	Sequential	Discrete	Multi
Taxi-Driving						



Task Environment-Examples

Environment	Observable	Deterministic	Static	Episodic	Discrete	# Agents
Chess	Fully	Deterministic	Semi	Sequential	Discrete	Multi
Poker	Partial	Stochastic	Static	Sequential	Discrete	Multi
Taxi-Driving	Partial	Stochastic	Dynamic	Sequential	Continuous	Multi



Task Environment-Examples

Environment	Observable	Deterministic	Static	Episodic	Discrete	# Agents
Chess	Fully	Deterministic	Semi	Sequential	Discrete	Multi
Poker	Partial	Stochastic	Static	Sequential	Discrete	Multi
Taxi-Driving	Partial	Stochastic	Dynamic	Sequential	Continuous	Multi
Medical-Diagnosis						



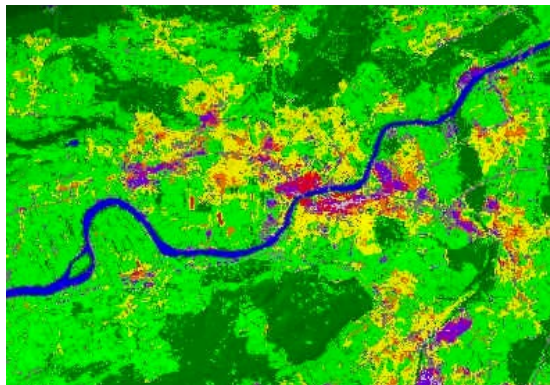
Task Environment-Examples

Environment	Observable	Deterministic	Static	Episodic	Discrete	# Agents
Chess	Fully	Deterministic	Semi	Sequential	Discrete	Multi
Poker	Partial	Stochastic	Static	Sequential	Discrete	Multi
Taxi-Driving	Partial	Stochastic	Dynamic	Sequential	Continuous	Multi
Medical-Diagnosis	Partial	Stochastic	Dynamic	Sequential	Continuous	Single



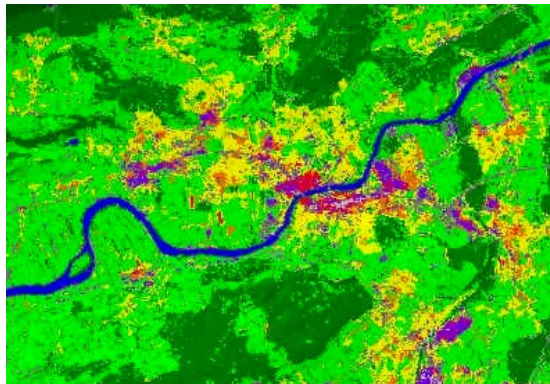
Task Environment-Examples

Environment	Observable	Deterministic	Static	Episodic	Discrete	# Agents
Chess	Fully	Deterministic	Semi	Sequential	Discrete	Multi
Poker	Partial	Stochastic	Static	Sequential	Discrete	Multi
Taxi-Driving	Partial	Stochastic	Dynamic	Sequential	Continuous	Multi
Medical-Diagnosis	Partial	Stochastic	Dynamic	Sequential	Continuous	Single
Image Analysis						



Task Environment-Examples

Environment	Observable	Deterministic	Static	Episodic	Discrete	# Agents
Chess	Fully	Deterministic	Semi	Sequential	Discrete	Multi
Poker	Partial	Stochastic	Static	Sequential	Discrete	Multi
Taxi-Driving	Partial	Stochastic	Dynamic	Sequential	Continuous	Multi
Medical-Diagnosis	Partial	Stochastic	Dynamic	Sequential	Continuous	Single
Image Analysis	Fully	Deterministic	Static	Episodic	Continuous	Single



Task Environment-Examples

Environment	Observable	Deterministic	Static	Episodic	Discrete	# Agents
Chess	Fully	Deterministic	Semi	Sequential	Discrete	Multi
Poker	Partial	Stochastic	Static	Sequential	Discrete	Multi
Taxi-Driving	Partial	Stochastic	Dynamic	Sequential	Continuous	Multi
Medical-Diagnosis	Partial	Stochastic	Dynamic	Sequential	Continuous	Single
Image Analysis	Fully	Deterministic	Dynamic	Episodic	Continuous	Single



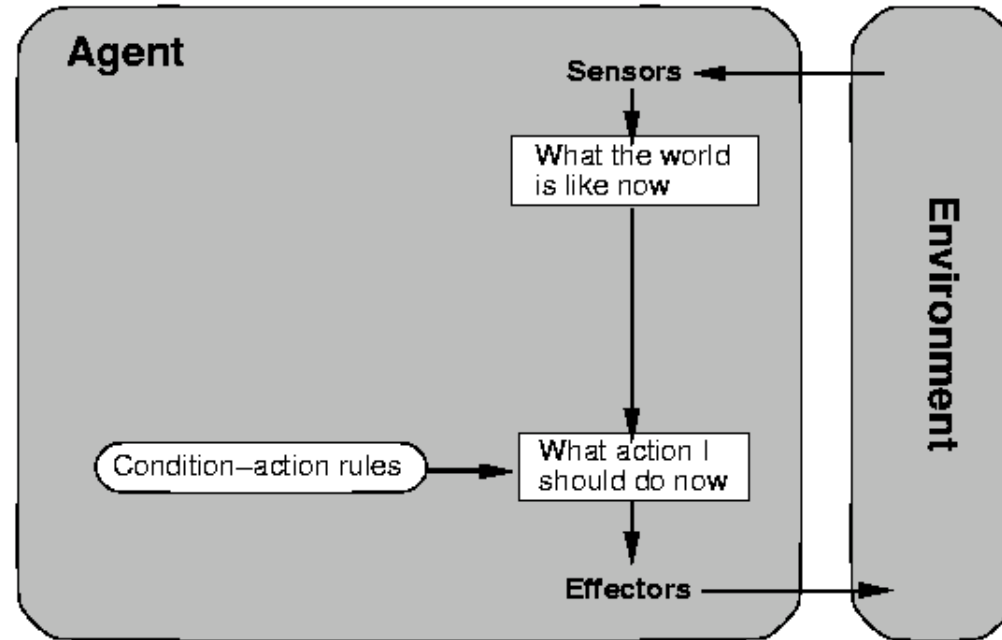
The real world is partially observable, stochastic,
dynamic and continuous
How do we handle it then?

Types of Agents

□ Types of agents (increasing in generality and ability to handle complex environments)

- Simple reflex agents
- Model based reflex agents
- Goal-based agents
- Utility-based agents
- Learning agents

Simple Reflex Agents



function SIMPLE-REFLEX-AGENT(*percept*) **returns** an action

persistent: *rules*, a set of condition-action rules

state $\leftarrow$ INTERPRET-INPUT(*percept*)

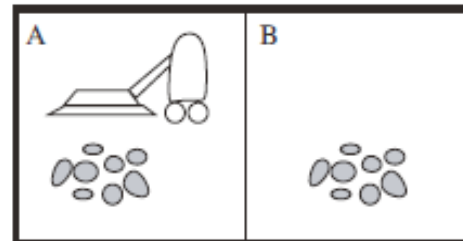
rule $\leftarrow$ RULE-MATCH(*state*, *rules*)

action $\leftarrow$ *rule*.ACTION

return *action*

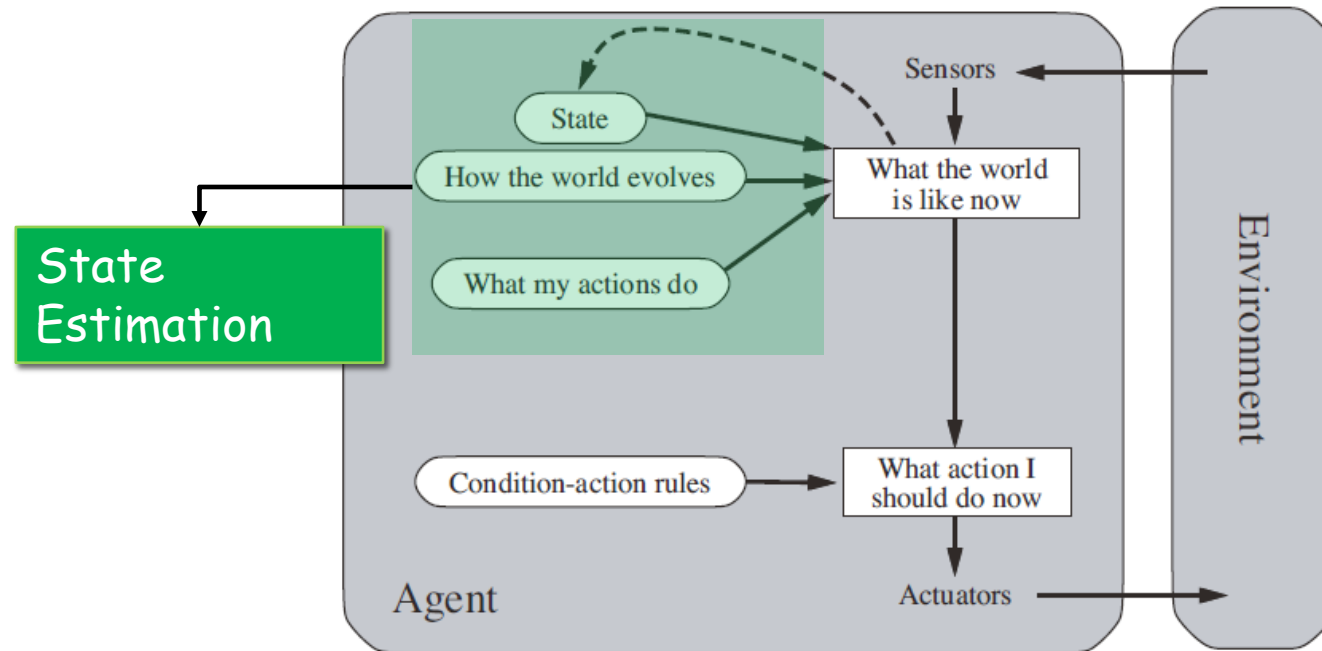
In what type of environments are simple reflex agents most likely to succeed?

Vacuum World



Percept sequence	Action
$[A, \textit{Clean}]$	<i>Right</i>
$[A, \textit{Dirty}]$	<i>Suck</i>
$[B, \textit{Clean}]$	<i>Left</i>
$[B, \textit{Dirty}]$	<i>Suck</i>
$[A, \textit{Clean}], [A, \textit{Clean}]$	<i>Right</i>
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$\vdots$	$\vdots$
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$\vdots$	$\vdots$

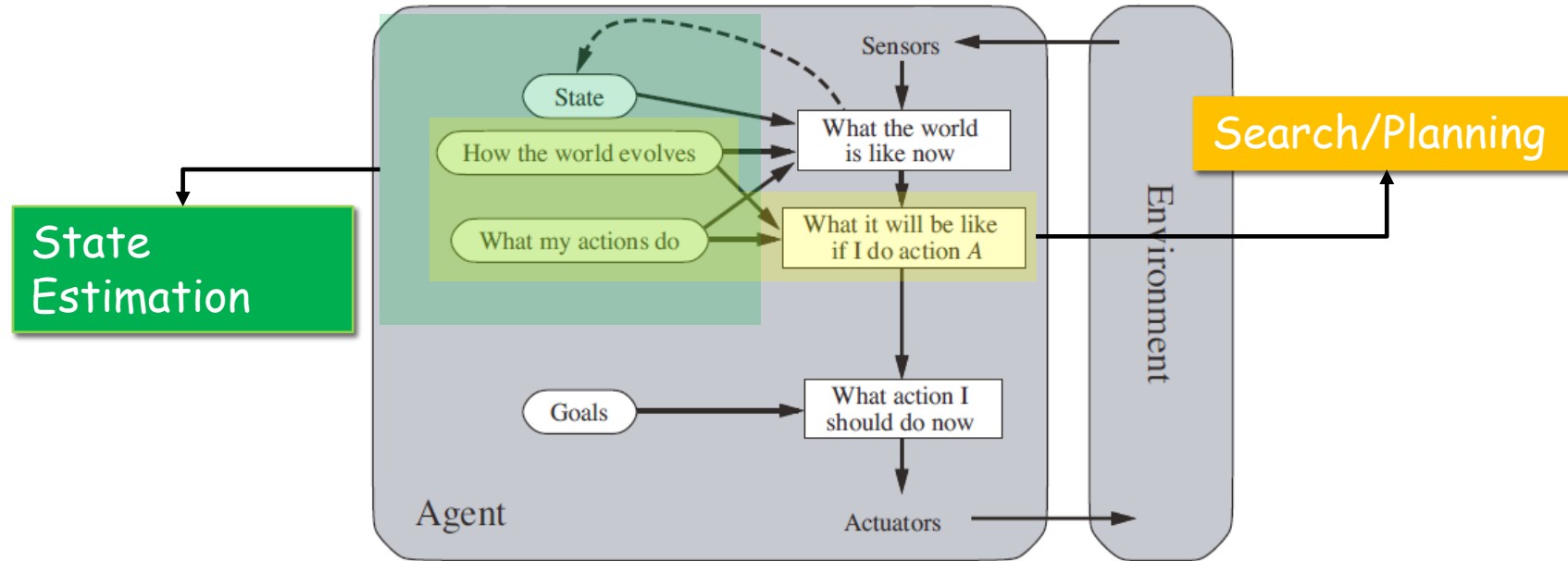
Model Based Reflex Agents



```
function MODEL-BASED-REFLEX-AGENT(percept) returns an action
  persistent: state, the agent's current conception of the world state
               model, a description of how the next state depends on current state and action
               rules, a set of condition-action rules
               action, the most recent action, initially none

  state ← UPDATE-STATE(state, action, percept, model)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

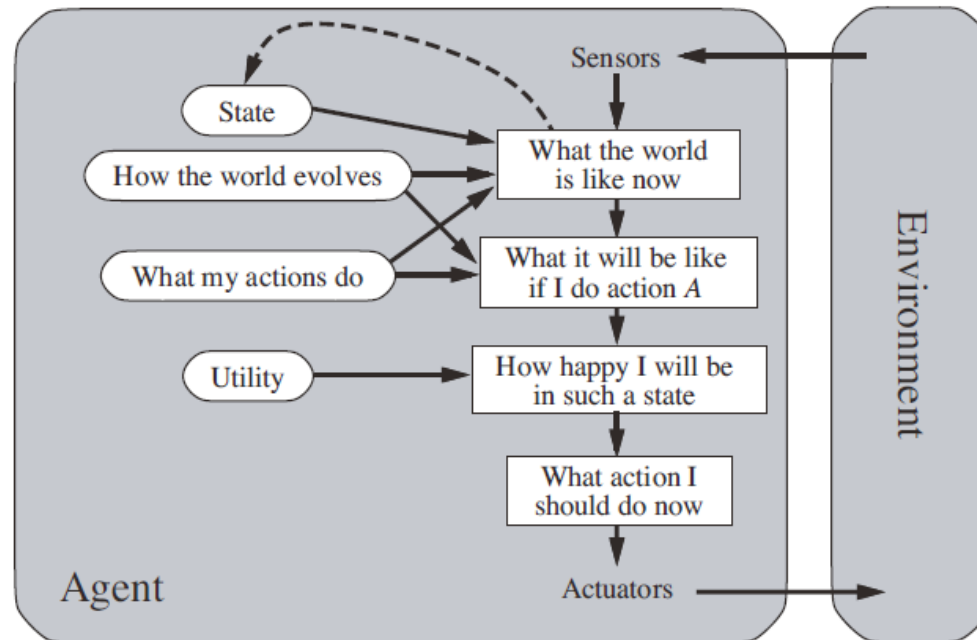

Goal Based Agents



Search: process of looking for a sequence of actions that reaches the goal state

Planning: can be viewed as search in a structured environment.

Utility Based Agents



- Utility function: internalization of the performance measure
- Conflicting goals
- Multiple uncertain goals
- Decision theoretic planning

Learning Agents

